

ET CRP AI HACKATHON 2026 | PHASE II SUBMISSION

TRINETRA

AI Digital Public Safety Intelligence Platform

Detailed Project Report

Problem Statement 6: AI for Digital Public Safety

Defeating Counterfeiting, Fraud & Digital Arrest Scams

Team DEDSEC

Theme: Smart Cities / Public Safety / Digital Trust / Geospatial Law Enforcement

Repository: <https://github.com/cybersd12321/trinetra-digital-public-safety>

Product tagline: Three eyes. One truth. (Citizen | Banks | Law Enforcement)

Document type: Working prototype project report for Phase II evaluation

Classification: Demo / Restricted | Synthetic operational data

1. Executive Summary

TRINETRA is an AI-powered Digital Public Safety Intelligence platform built by Team DEDSEC for the Economic Times CRP AI Hackathon 2026 (Problem Statement 6). It equips law enforcement agencies (LEA), financial institutions, and citizens with proactive tools to detect, disrupt, and respond to digital fraud networks, counterfeit currency circulation, and organised digital arrest scam operations.

The platform shifts the operating doctrine from reactive case investigation (FIR after victimisation) to predictive threat neutralisation at the point of contact (before fund transfer completes). Six live modules form a multi-source intelligence fabric: Command Centre, Digital Arrest Scam Detector, Counterfeit Currency Agent, Fraud Network Graph, Geospatial Crime Intelligence, and Citizen Fraud Shield.

Key demo metrics: Rs 18.4 Cr loss prevented (simulated window), 6.4 hour average lead time before mass victimisation, 1.8% citizen false-positive rate, 19 fraud networks mapped, fully offline-capable intelligence engines for reliable demos.

2. Problem Statement Context

2.1 Official challenge (PS-6)

Build an AI-powered Digital Public Safety Intelligence platform that equips law enforcement agencies, financial institutions, and citizens with proactive tools to detect, disrupt, and respond to digital fraud networks, counterfeit currency circulation, and organised scam operations, shifting from reactive case investigation to predictive threat neutralisation.

2.2 Problem scale (from problem brief)

- India registered 1.14 million cybercrime complaints in 2023 (+60% YoY).
- Digital arrest scams (impersonating CBI, ED, Customs over multi-day video coercion) defrauded citizens of over Rs 1,776 crore in nine months of 2024 (Ministry of Home Affairs reporting context).
- These are industrialised operations run from fraud compounds, often across borders, using spoofed numbers, AI-generated voices, and fake government portals.
- Counterfeit currency remains a persistent threat: RBI Annual Report 2025 context flags record FICN (Fake Indian Currency Notes) seizures, including high-quality Rs 500 fakes that can defeat manual detection in routine banking operations.

2.3 The intelligence gap

What law enforcement lacks is not evidence after the fact. It is intelligence before mass victimisation occurs, and reliable tools at the point of contact rather than the point of complaint. Solving this requires convergence of:

- Financial transaction intelligence
- Communication network analysis
- Physical security (counterfeit detection)
- Real-time public safety coordination

This is exactly the multi-source, multi-agency intelligence problem where AI can be transformative. TRINETRA is designed as that fusion layer.

3. Objectives and Scope

3.1 Primary objectives

- Detect digital arrest and related scam scripts in real time with explainable scores.
- Authenticate currency notes via multi-feature analysis for banks and field officers.
- Map coordinated fraud campaigns as graph intelligence packages across jurisdictions.
- Prioritise geospatial hotspots for patrol and inter-district sharing.
- Provide a citizen-facing conversational shield with 1930 / NCRB reporting guidance.
- Present a unified LEA Command Centre for multi-agency situational awareness.

3.2 In-scope (Phase II prototype)

- Working web prototype with six interactive modules and public GitHub documentation.
- Explainable, offline-first AI engines (no mandatory cloud API key for demo).
- Synthetic but realistic incidents, networks, hotspots, and sample inputs.
- Architecture description, pitch deck, and this project report.

3.3 Out of scope (Phase II) / roadmap

- Live telecom session intercept or production bank FRA integrations.
- WhatsApp Business / IVR multi-language channels (designed; web demo live).
- Production court evidence vault with full chain-of-custody hardware.
- Trained deep CV models on real FICN image corpora (feature schema is ready).

4. Proposed Solution: TRINETRA

TRINETRA (Sanskrit: three eyes) is a multi-tenant intelligence platform with one shared core of engines and three stakeholder surfaces: citizen, bank/POS, and LEA. Every high-risk verdict is designed to be explainable (signals and features) so packages remain auditable for operational and legal use.

4.1 Platform modules

Module	Primary user	Function
Command Centre	LEA / MHA ops	Fused KPIs, charts, live incidents
Scam Detector	Cyber cell / telecom	NLP risk score + MHA alert draft
Currency Agent	Bank / POS / field	Multi-feature note authenticity
Fraud Network Graph	Investigators	Entity graph + intel packages
Geospatial Intel	District command	Hotspots and patrol priority
Citizen Shield	Public	Chat risk verdict + 1930/NCRB SOP

4.2 Doctrine: Ingest, Score, Fuse, Act

- Ingest: Call scripts, SMS, note scan features, transaction linkages, citizen reports at the moment of contact.
- Score: Explainable engines fire weighted linguistic signals or optical features with confidence.
- Fuse: Graph and geospatial layers stitch entities across districts so compounds do not appear as isolated FIRs.
- Act: MHA-style alert drafts, 1930 paths, patrol priority, and bank FRA handoff while the session may still be live.

5. How the Prototype Works

5.1 Digital Arrest Scam Detector

Users paste a call transcript, SMS, or chat message. A weighted multi-pattern NLP engine scans for industrial scam signatures and produces a 0 to 100 risk score, risk level, category, confidence, explainable signal list, recommended actions, and (when high risk) an MHA/cyber alert draft that can be copied for handoff.

Signal categories (examples)

- Law enforcement impersonation (CBI, ED, Customs, cyber cell language)
- Arrest / warrant coercion and isolation instructions (do not hang up / tell family)
- Framing for money laundering, drugs, Aadhaar misuse
- Urgent transfer demands and court-linked / verification accounts
- Multi-hour video hostage patterns
- Fake KYC urgency, spoofed domains, OTP harvest, investment guarantees

Safe markers (e.g., official bank language telling customers never to share OTP) discount the score to reduce citizen false positives. Demo samples include a critical digital arrest script versus a safe bank credit SMS for contrast.

5.2 Counterfeit Currency Agent

The Currency Agent accepts a structured note-scan feature vector (proxy for a computer-vision capture pipeline on mobile, POS, or counting machines). Features include microprint, security thread, watermark, see-through register, UV response, print noise, edge sharpness, bleed-through, and serial pattern.

Each feature contributes a weighted pass/fail/uncertain score. Consensus confidence classifies the note as genuine, suspect, or counterfeit, and returns field-officer guidance (accept, quarantine and re-verify, or seize and escalate). Production path: camera/UV capture models output the same feature schema into the same scorer.

5.3 Fraud Network Graph Intelligence

An interactive force-layout graph visualises a demo cluster (Compound Alpha): compound, handlers, spoofed numbers, devices, money mules, aggregation accounts, crypto off-ramp, and victims. Clicking a node surfaces risk, metadata, and linked relations. Intelligence packages attach estimated loss, jurisdictions, lead agency, status (active / monitoring / disrupted), and a narrative suitable for multi-agency handoff and court-oriented packaging.

5.4 Geospatial Crime Pattern Intelligence

City-level hotspots across India are plotted with crime type (digital arrest, counterfeit, UPI fraud, mixed), intensity, 30-day incident counts, and trend (up / down / stable). Selecting a hotspot ranks patrol priority and suggests inter-district intelligence sharing. This supports command-centre playbooks: Detect, Prioritise, Share.

5.5 Citizen Fraud Shield

A conversational web channel (parity roadmap for WhatsApp and IVR) walks citizens through risk assessment. It wraps the same scam engine, explains signals in plain language, and guides reporting via helpline 1930 and cybercrime.gov.in (NCRB). Hard rules are reinforced: never share OTP/UPI PIN; police/CBI/ED never demand money on video calls.

5.6 Command Centre

The LEA dashboard fuses demo KPIs (active threats, loss prevented, sessions flagged, citizen alerts), prevented-versus-reported loss trends, threat-mix composition, and a live incident feed with severity and status. It is the multi-agency ops surface for situational awareness during evaluation and demo walkthroughs.

6. System Architecture

6.1 Logical layers

- Presentation: Next.js App Router (landing + six modules), responsive ops UI.
- Application: Client-side orchestration and demo state.
- Intelligence: Pure TypeScript engines (scam, currency, shield, graph layout, geo scoring).
- Data: Synthetic incidents, clusters, hotspots, and sample inputs.
- Integration adapters (roadmap): NCRB, 1930, bank FRA, telecom session flags, state CERTs.

6.2 Architecture diagram (textual)



6.3 Design principles

- Point-of-contact intelligence: decide before money moves.
- Explainability: every score exposes signals/features for auditability.
- Multi-tenant surfaces: citizen UX differs from LEA ops UX; shared engines.
- Offline-demo reliability: engines run without cloud keys.
- Audit trail: alert drafts and packages are exportable text artifacts.

7. Technology Stack

Layer	Technology	Role
Framework	Next.js 16 (App Router)	Routing, SSR/SSG, app structure
Language	TypeScript	Type-safe engines and UI
UI library	React 19	Interactive modules
Styling	Tailwind CSS v4	Design system and layout
Charts	Recharts	Command Centre analytics
Icons	Lucide React	Consistent ops iconography
Motion	CSS transforms / opacity	Radar, pulse (reduced-motion safe)
Graph / Map	Custom SVG	No heavy vendor lock-in for demo
AI (prototype)	Deterministic TS engines	Explainable offline scoring
AI (roadmap)	SpaceXAI / xAI grok-4.5	Multilingual chat, multimodal OCR

Layer	Technology	Role
VCS / collab	Git + GitHub (public)	Phase II working prototype host
Package mgr	npm	Dependency management

7.1 Alignment with suggested technologies (PS-6)

Suggested (brief)	TRINETRA coverage
NLP / LLMs	Weighted multi-pattern NLP scam classifier + Shield chat
Computer Vision	Currency multi-feature agent (CV pipeline schema ready)
Graph AI & network analysis	Fraud network graph + cluster packages
Geospatial intelligence	Hotspot map, intensity, patrol priority
Speech AI	Roadmap: voice spoof / deepfake detection
Agentic multi-source fusion	Command Centre fusion of all sensors

8. Evaluation Focus Mapping (PS-6)

The problem brief emphasises counterfeit detection quality, digital arrest precision/recall, fraud network lead time before mass victimisation, very low false-positive rate for citizen tools, and auditability of intelligence packages.

Evaluation focus	How TRINETRA addresses it
Counterfeit accuracy	8+ feature checks; genuine/suspect/counterfeit consensus
Digital arrest P/R	Weighted signal fusion; explainable evidence per hit
Network lead time	Graph clustering before mass victimisation (demo 6.4h)
Citizen false positives	Safe markers + conservative thresholds (demo 1.8% FP)
Legal auditability	Exportable packages: jurisdictions, lead, entity graph

8.1 Judging criteria alignment

Criterion	Weight	Our emphasis
Innovation	25%	Point-of-contact fusion of five sensor types + citizen
Business impact	25%	Loss prevention, multi-stakeholder value, MHA path
Technical excellence	20%	Typed engines, explainability, architecture
Scalability	15%	Stateless scoring; graph DB / edge roadmap
User experience	15%	Distinct ops UI + simple citizen shield chat

9. Impact and Stakeholders

- Citizens: Instant risk verdicts; fewer multi-day digital arrest hostage situations; guided 1930 / NCRB reporting.
- Banks and POS: Counterfeit agent plus mule graph context lowers fraud loss and speeds FRA action.
- Telecom: Session-level flags before UPI drain completes (integration roadmap).
- LEA / MHA: Cross-jurisdiction packages, geospatial patrol priority, shared ops picture.
- India (systemic): Move from crores lost after FIR to rupees saved before transfer.

10. Scalability, Security, and Roadmap

10.1 Scalability approach

- Citizen chat: Stateless scoring; horizontal scale on edge/workers.
- Bank POS: On-device feature extraction + light server-side score.
- LEA graph: Graph database (e.g., Neo4j) + streaming link ingestion.
- National geo: Tile service with state partitions.

10.2 Security and privacy (production design)

- PII minimisation on citizen channel; RBAC evidence vault for LEA.
- Immutable hashes of intelligence packages for chain of custody.
- No raw OTP/PIN storage; redaction in logs.

10.3 Roadmap

- Phase II (now): Working multi-module prototype, public GitHub, deck, report.
- Pilot (90 days): Bank POS + cyber cell pilot; WhatsApp Shield; live FRA feed.
- Scale: State CERT mesh; 12-language IVR; voice deepfake detection; GNN mule discovery.

11. Phase II Deliverables Checklist

Required deliverable	Status	Location
Working prototype	Complete	GitHub public repository
Architecture diagram	Complete	README + Section 6 of this report
Presentation deck	Complete	presentation/TRINETRA_ET_Hackathon_2026.pptx
Demo video (3-4 min)	Team records	Walkthrough of all six modules
Detailed project PDF	Complete	This document

11.1 How to run the prototype

```
git clone https://github.com/cybersd12321/trinetra-digital-public-safety.git
cd trinetra-digital-public-safety
npm install
npm run dev
# open http://localhost:3000
# no API keys required for Phase II demo engines
```

11.2 Module routes

URL path	Module
/	Landing / product thesis
/dashboard	Command Centre
/scam-detector	Digital Arrest Scam Detector
/counterfeit	Counterfeit Currency Agent

URL path	Module
/network	Fraud Network Graph
/geospatial	Geospatial Crime Intelligence
/citizen	Citizen Fraud Shield

12. Repository Structure (summary)

trinetra/	
src/app/	Landing + module routes
src/components/	Shell, graph, map, landing radar, UI atoms
src/lib/engines/	scamDetector, currencyDetector, shieldAgent, graphLayout
src/lib/data/	incidents, network, hotspots, samples
docs/	ARCHITECTURE.md, this project report PDF
presentation/	Pitch deck (PPTX)
README.md	Setup, architecture, evaluation alignment

13. Conclusion

Digital arrest compounds, high-grade counterfeit notes, and mule networks industrialise faster than reactive FIRs can respond. TRINETRA, built by Team DEDSEC for ET AI Hackathon 2026 Problem Statement 6, demonstrates a practical, explainable, multi-source AI platform that places intelligence at the point of contact for citizens, banks, and law enforcement.

The Phase II prototype is fully runnable from a public GitHub repository, covers the illustrative solution areas in the problem brief (scam detection, counterfeit identification, fraud graph intelligence, geospatial crime patterns, and multi-channel citizen shield), and is structured for evaluation on innovation, impact, technical excellence, scalability, and user experience.

Three eyes. One truth. Stop the transfer, not just file the FIR. | Team DEDSEC

References and sources (problem context)

- ET CRP AI Hackathon 2026 Problem Statements PDF: PS-6 AI for Digital Public Safety.
- Contextual statistics as stated in the official problem brief (cybercrime volume, digital arrest loss figures, FICN / RBI reporting references).
- Helpline 1930 (Citizen Financial Cyber Fraud Reporting) and NCRB portal cybercrime.gov.in as official citizen reporting channels referenced in product UX.
- Prototype repository: <https://github.com/cybersd12321/trinetra-digital-public-safety>